



AI ENGINEERING TRACK · MODULE 2 PROJECT

Churn Prediction for Wasla Telecom

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The Client Brief

Wasla Telecom (وصلة — “connection”) is a mid-size telecom operator serving customers across the region. Over the past year, Wasla lost roughly 27% of its subscriber base to churn — customers cancelling their plans and switching to a competitor. Every cancelled contract is lost recurring revenue, and by the time a customer actually cancels, it's already too late to win them back.

Wasla's VP of Customer Retention, Mona Adel, has brought your team in as data science consultants with one mission:

“Tell us which customers are at risk of leaving before they leave — and tell us why — so our retention team can step in early.”

This is a real, well-studied dataset of actual customer records (not simulated), so the patterns you find are genuine business signal, not manufactured examples.

Your Task

Step 1 — Explore the Records (EDA)

Understand who's churning and look for patterns across contract type, tenure, billing, and add-on services. Pay close attention to data types — one column in particular hides a bug that only shows up if you check carefully.

Step 2 — Clean & Preprocess

Fix the hidden data-type issue, encode categorical fields consistently, and scale where needed.

Step 3 — Train Multiple Models

Train Logistic Regression, Random Forest, and XGBoost, and compare them fairly using cross-validation.

Step 4 — Select the Best Model

Choose an evaluation metric that actually matters here — remember, only about 1 in 4 customers churn, so accuracy alone can be misleading.

Step 5 — Write the Evaluation Report

Turn your model into a set of practical recommendations Wasla's retention team can act on — e.g., “customers on month-to-month contracts with no tech support are far more likely to churn — prioritize outreach to this group first.”

The Dataset

File: wasla_customer_records.csv · 7,043 customer records · 21 fields

Category	Fields
Demographics	gender, SeniorCitizen, Partner, Dependents
Account details	tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Services subscribed	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Target	Churn (Yes / No) — whether the customer left Wasla

These are Wasla's real customer records, provided exactly as the retention team pulled them from the billing system — quirks and all.

Deliverables

1. **One notebook** — containing your EDA, preprocessing, all three trained models, and your model comparison.
2. **An evaluation report for Mona Adel** — a short, non-technical summary translating your best model's findings into concrete retention actions Wasla can take.

Due Date

By the start of Session 9. Most of this will already be built together in class today — this is mainly about polishing your notebook and writing the report for the client.

A Note on Approach

Mona doesn't care which algorithm you used — she cares whether her team can act on it. A slightly less accurate model with a clear, actionable report beats a marginally better model with no business takeaway.

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